Adaptive Credit-Weighted Online Mirror Descent for Sparse Verifiable Reward Reinforcement Learning

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Abstract

Sparse terminal rewards make it difficult to decide which actions in a long trajectory deserve a large policy update. Broadcasting one group-relative advantage to every decision is inexpensive, but finite-sample correlations can move policy mass at decisions that do not affect the reward. We study a controlled sequence setting in which the reward-relevant positions are known for evaluation but hidden from the optimizer. We introduce Online RC-OMD, a low-cost reliability-calibrated variant of Online Mirror Descent (OMD). It tracks exponentially weighted first and second moments of group-relative action-score estimates and uses their persistent signal-to-uncertainty ratio to scale the local KL-mirror step. The method requires one score computation per group and $O(H|\mathcal{A}|)$ persistent state. In a frozen out-of-distribution protocol covering four threshold-reward tasks and ten seeds per task, Online RC-OMD remained within 0.0018 normalized success AUC of a tuned uniform Group OMD baseline while using only 18.9%–24.8% as much absolute cumulative KL at known distractor positions. These results support a narrow Pareto claim about controlled policy movement; they do not establish causal credit assignment or performance under function approximation.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) supplies an attractive training signal for tasks whose final outcomes can be checked automatically. The reward is often binary and observed only after a complete trajectory. A group-relative baseline can avoid learning a separate critic, as in Group Relative Policy Optimization (GRPO) [3], but the same trajectory-level advantage is then commonly propagated through many decisions. This leaves an optimizer-level question: when the available credit signal is a noisy proxy, how much policy movement should each decision be allowed to make?

Recent work has attacked token-level credit assignment using entropy-aware weights and related local signals [2, 5]. Entropy can be informative, but uncertainty is not identical to reward relevance. A decision can be uncertain and irrelevant, or confident and pivotal. We therefore treat credit reliability as a quantity to be estimated rather than equating it with entropy. Our approach is deliberately smaller than language-model RL: a factorized tabular policy lets us observe local policy movement exactly and construct environments with known distractor decisions.

Our contributions are:

1. We construct controlled sparse terminal-reward sequence tasks with exact counterfactual credit and known reward-irrelevant positions for evaluation. The suite includes settings where policy entropy is aligned with, and deliberately misleading about, true positional importance.

2. We propose Online RC-OMD, which converts persistent agreement in online group-relative action scores into per-position OMD step scales. Unlike a per-group bootstrap estimator, it adds no rollout resampling and maintains only $O(H|\mathcal{A}|)$ state.

3. We report development results separately from a protocol frozen before evaluation. The pre-registered comparison passes all four unseen scenarios under a declared AUC–distractor-KL rule.

The central empirical claim is a trade-off, not universal dominance: reliability calibration directs substantially less policy movement toward known distractors while retaining almost all learning efficiency in the tested controlled tasks.

2 Related Work

Mirror-descent policy optimization. Mirror descent replaces Euclidean proximity with a Bregman divergence and is particularly natural on a probability simplex [1]. MDPO connects this geometry to practical trust-region policy optimization by combining a linearized RL objective with a proximity term [4]. We use the negative-entropy mirror map, for which the proximity term is KL and the exact local update is exponentiated gradient.

Group-relative optimization and sparse verifiable rewards. DeepSeekMath introduced GRPO as a critic-free relative policy optimization method for mathematical reasoning [3]. Outcome-only feedback saves the cost of a process reward model, but it leaves temporal credit coarse. A recent survey organizes the rapidly growing LLM credit-assignment literature by granularity and methodology [6].

Entropy-aware credit. EAPO analyzes reward polarity and token entropy, and shows that entropy bounds the credit a token can carry under its information-theoretic formulation [2]. ACPO uses a mode-local surrogate entropy to modulate token updates asymmetrically [5]. Our study does not compete with these systems at language-model scale. Instead, it tests a distinct diagnostic hypothesis: persistent score agreement can regulate local mirror geometry even when instantaneous entropy is a misleading proxy for reward relevance.

3 Controlled Problem

Let a trajectory be $\tau = (a_1, \ldots, a_H)$ with $a_k \in \mathcal{A} = \{1, \ldots, A\}$. The factorized policy is $\pi(\tau) = \prod_{k=1}^{H} \pi_k(a_k)$. A hidden set $\mathcal{K} \subseteq \{1, \ldots, H\}$ contains the pivotal positions. Each $k \in \mathcal{K}$ has a target action $a_k^\star$. The terminal reward is

$$R(\tau) = \mathbf{1}\left\{\sum_{k\in\mathcal{K}} \mathbf{1}\{a_k = a_k^\star\} \geq m\right\}. \tag{1}$$

Positions outside $\mathcal{K}$ are distractors: changing them never changes the reward. The special case $m = |\mathcal{K}|$ is an all-match or "needle" task. Threshold tasks use $m < |\mathcal{K}|$.

For each update, K trajectories are sampled and assigned group-relative advantages

$$\widehat{A}_i = R(\tau_i) - \frac{1}{K}\sum_{j=1}^{K} R(\tau_j). \tag{2}$$

We estimate the gain for action a at position k with inverse propensity weighting,

$$\widehat{g}_{t,k,a} = \frac{1}{K}\sum_{i=1}^{K} \frac{\mathbf{1}\{a_{i,k} = a\}\widehat{A}_i}{\max(\pi_{t,k}(a), p_{\min})}, \tag{3}$$

with a fixed importance cap in the implementation. Uniform Group OMD applies this noisy action score at every position. Although the expected distractor score is zero in our controlled environment, its finite-group realization need not be, so repeated updates may accumulate irrelevant KL drift.

For diagnosis only, the known transition and reward structure gives exact on-trajectory counterfactual credit

$$c_k(\tau, \pi) = Q^\pi(s_k, a_k) - V^\pi(s_k). \tag{4}$$

The optimizer never receives the pivotal set or oracle credit unless explicitly labelled as an oracle baseline.

4 Reliability-Calibrated OMD

4.1 Uniform KL-mirror update

At position k, negative-entropy OMD solves

$$\pi_{t+1,k} = \arg\max_{p\in\Delta_A} \left\{\eta\langle \widehat{g}_{t,k}, p\rangle - D_{\mathrm{KL}}(p \,\|\, \pi_{t,k})\right\}, \tag{5}$$

with solution

$$\pi_{t+1,k}(a) = \frac{\pi_{t,k}(a)\exp(\eta\widehat{g}_{t,k,a})}{\sum_b \pi_{t,k}(b)\exp(\eta\widehat{g}_{t,k,b})}. \tag{6}$$

Table 1: Frozen OOD comparison. AUC difference and KL ratio are Online minus Uniform and Online divided by Uniform, respectively. Values are means across ten seeds; the seed is the replication unit.

Scenario	Uniform AUC	Online AUC	ΔAUC	KL ratio	Runtime ratio
Dense 2-of-6	0.992182	0.990865	-0.001317	0.248	1.107
Needle 5-of-5	0.980026	0.980782	0.000755	0.239	1.081
Threshold 3-of-5	0.992195	0.990567	-0.001628	0.189	1.069
Threshold 4-of-6	0.989446	0.987688	-0.001758	0.199	1.079

4.2 Online reliability estimate

We center each score vector across actions and maintain exponentially weighted first and second moments with decay β. Let $\widehat{\mu}_{t,k}$ be the running mean vector, $\widehat{s}_{t,k}$ its estimated standard-error vector, and $n_{\text{eff},t}$ the effective sample size of the exponential weights. We define

$$q_{t,k} = \min\left\{1, \frac{n_{\text{eff},t}}{n_{\text{warm}}}\right\}\left[1 - z\frac{\|\widehat{s}_{t,k}\|_2}{\max(\|\widehat{\mu}_{t,k}\|_2, \epsilon)}\right]_+, \tag{7}$$

$$\alpha_{t,k} = \alpha_{\min} + (1-\alpha_{\min})q_{t,k}. \tag{8}$$

The first factor prevents confidence before enough effective observations; the second rewards a persistent signal relative to its uncertainty. The local OMD update replaces η in Eq. (6) by $\eta\alpha_{t,k}$. The current score determines direction, whereas the historical reliability estimate controls how far the policy may move.

Proposition 1 (Exact local movement control). *For any actions a, b with positive probability, the reliability-calibrated update satisfies*

$$\log\frac{\pi_{t+1,k}(a)}{\pi_{t+1,k}(b)} - \log\frac{\pi_{t,k}(a)}{\pi_{t,k}(b)} = \eta\alpha_{t,k}\big(\widehat{g}_{t,k,a} - \widehat{g}_{t,k,b}\big). \tag{9}$$

Thus reliability scales every local log-odds displacement linearly, without changing the ranking induced by the current action-score vector.

Proof. Take the log ratio of the two normalized exponentiated-gradient probabilities; the shared normalizer cancels. □

The estimator uses one action-score computation per group. Its first and second moments require $O(HA)$ memory and $O(HA)$ arithmetic per update, matching the asymptotic score-update cost of the uniform baseline.

5 Experimental Design

We separate exploratory development from frozen evaluation. Development tasks were used to construct entropy counterexamples, compare bootstrap and online reliability, and choose the online configuration. The subsequent OOD protocol, including tasks, seeds, metrics, parameter values, and decision rule, was committed before execution under identifier `ood-v1-2026-08-08`.

The frozen comparison used ten seeds per task. Uniform Group OMD used base step 0.75. Online RC-OMD used base step 1.25, decay 0.9, $z = 1$, warm-up 8, and reliability floor 0.1. These unequal base steps were selected on development tasks to compare a Pareto-matched operating point; a separate same-step diagnostic was also frozen. Normalized success AUC is the mean exact success probability across evaluation checkpoints. Distractor KL is the sum over training iterations and known distractor positions of the absolute local KL movement.

A scenario passed when Online RC-OMD was no more than 0.01 below Uniform in success AUC and used at most 50% of Uniform's distractor KL. The method-level Go decision required at least three of four scenario passes. Runtime feasibility required an Online-to-Uniform ratio below 1.5 in at least three scenarios.

6 Results

6.1 Pre-registered OOD evaluation

All four scenarios passed the pre-declared rule (Table 1), yielding a Go decision. Online RC-OMD retained AUC to within 0.0018 while using between 18.9% and 24.8% as much distractor KL. Runtime overhead was 6.9%–10.7%,

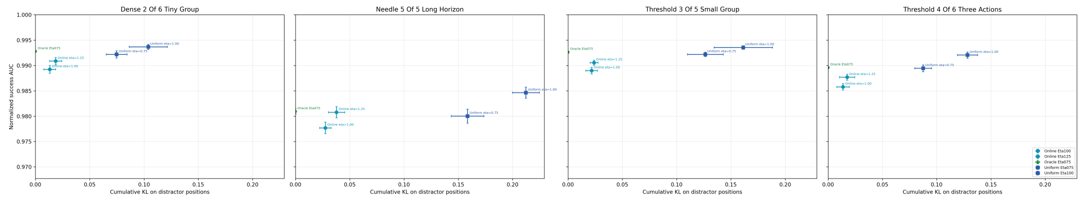

Figure 1: Pre-registered OOD success–distractor-drift comparison. The generated artifact is reproduced from the frozen configuration; the numerical decision is based on the machine-readable summary rather than the figure.

so all four scenarios also passed the systems rule. The scenarios changed reward structure, horizon, group size, initial success probability, and included both two- and three-action policies.

At a common base step of 1.0, Online RC-OMD had 0.0044–0.0069 lower AUC, but used only 10.7%–13.4% as much distractor KL. This diagnostic confirms that the main result is an update-allocation trade-off, not evidence that reliability scaling always raises reward at an identical nominal step size.

6.2 Development diagnostics

In entropy-aligned development tasks, entropy weighting improved early learning; when distractor positions were initialized with high entropy and pivotal positions with lower entropy, the same heuristic reduced sample efficiency. This controlled reversal motivated reliability estimation. Per-group bootstrap reliability reduced distractor drift but cost roughly 7–9× the uniform runtime. Replacing bootstrap resampling with online moments reduced runtime to within roughly 7%–13% of Uniform on the development tasks. These measurements are exploratory and are not pooled with the frozen OOD result.

7 Discussion and Limitations

The results are consistent with a simple mechanism: reward-relevant score directions recur, whereas finite-sample distractor directions fluctuate, so running agreement can allocate a larger KL budget to the former. This is an associational mechanism, not a causal-credit theorem. A persistent spurious correlation can also look reliable, and estimator lag can suppress newly useful directions after a policy or task change.

The current scope has five important limits. First, the policy is tabular and factorized; shared neural parameters may couple local updates. Second, known distractor positions are used only to evaluate drift, an oracle unavailable in real RLVR. Third, the principal methods use base steps chosen on development tasks. Fourth, ten seeds quantify simulation variation but not universality. Fifth, no language model, process reward model, or long natural-language trajectory has been tested. These limitations make function approximation the next necessary experiment, while a language-model study remains optional.

8 Conclusion

We studied whether the reliability of a noisy step-level credit proxy should control local mirror-descent geometry under sparse terminal rewards. A low-cost online moment estimator produced a reproducible Pareto improvement in four frozen controlled OOD scenarios: nearly unchanged success learning with substantially less policy movement at known distractor decisions. The next stage must determine whether this behavior survives shared-parameter function approximation, where local policy constraints are only approximately realizable.

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